

ABC PUBLIC SCHOOL

Subject: Science

Class: 10

Time Allowed: 45 minutes

Maximum Marks: 50

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Section: _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
 2. Write your answers in the space provided.
 3. Read each question carefully before answering.
 4. Marks are indicated against each question.
- Attempt all questions. Write clearly and concisely.

SECTION A

Short Answer Questions

Attempt all questions

1. What is the main function of chlorophyll in a plant leaf? (2)
2. Name the primary sugar molecule produced at the end of photosynthesis. (2)
3. Define the overall process of photosynthesis. (2)

SECTION B

Long Answer Questions

Attempt all questions

1. Distinguish between the light reaction and the dark reaction of photosynthesis based on their energy requirements. (3)
2. Explain the role of sunlight in initiating the light-dependent reactions. (3)
3. Where exactly within the chloroplast do the light reaction and dark reaction occur? (3)
4. Describe the fate of the glucose molecules formed during photosynthesis within the plant. (3)
5. Identify two crucial products of the light reaction that are then utilized in the dark reaction. (3)
6. Why is the dark reaction also referred to as the light-independent reaction, even though it indirectly depends on light? (3)
7. Illustrate how plants obtain the raw materials necessary for the synthesis of glucose. (3)
8. Compare the primary outcomes of the light reaction with those of the dark reaction. (3)

SECTION C

Analytical Questions

Attempt all questions

1. Analyze the critical importance of the continuous regeneration of ADP and NADP⁺ for the sustained process of photosynthesis. What would happen if this regeneration failed? **(5)**
2. Discuss how various environmental factors, such as light intensity and water availability, can significantly influence the rate of glucose formation in a plant. **(5)**
3. Evaluate the statement: 'Photosynthesis is the cornerstone of Earth's ecosystem.' Provide specific examples to support your evaluation, considering both plant and animal life. **(5)**
4. Design a conceptual experiment to demonstrate that carbon dioxide is essential for the dark reaction of photosynthesis, even in the presence of light. Outline your approach and expected observations. **(5)**

Best of Luck!